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THE

PHILOSOPHY OF EVOLUTION

TOGETHER WITH A PRELIMINARY ESSAY ON

THE METAPHYSICAL BASIS OF SCIENCE.

TWO PAPERS

Read before The Wisconsin Academy of Sciences, Arts, and Letters at the Annual
Meetings of February, 1873 and February, 1874.

BY

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THE METAPHYSICAL BASIS OF SCIENCE.

All knowledge is essentially one. The object-matter upon which intellect exerts itself, does not affect the subjective act of knowing. Physics, when stripped of that which is merely contingent, becomes metaphysics. Physical science deals with object-matter, and discusses the signs by which nature communicates her message—that is, phenomena. Metaphysical science has to do with the subject-mind, and discusses the meaning of the message. The one converts God's hieroglyphics into easily-intelligible language; the other translates this language into Idea. If this be true, there must be a unity of method in all science, however great the diversity of the object-matter investigated. This method is subjectively determined, that is, by the constitution of the mind, and not by the particular form of matter upon which intellectual energy may be exerted. If there is an essential unity in all knowledge, it is because there is a corresponding unity of method in all mental activity. It is only when we look upon what is to be known, that truth separates into sciences; but particular truths become particular sciences only under assumed relations to the whole of which they form a part.

Objectively considered, science is classified knowledge; subjectively viewed, it is the laws or principles according to which knowledge is classified. Every actor implies an act—every thinker a thought. We may therefore universally make this dual classification, according as we view the mental operation involved, or the attributes of objects which form the subject of thought. The possibility of science is conditioned upon the possibility of classification. Mere knowledge is not science, as the world ought to have learned by costly experience. Even classified knowledge may not be science; it becomes science not through previous classification, but in the act of being classified, and therefore only as the principle of classification is apprehended—that is, only as the particular application of the law of generalization is distinctly recognized. A man may know a book and know nothing more; he knows the science only when he is capable of making the book for himself. Mere knowledge thus differs from science in that the one is held only by the apprehensive powers of the mind, while the other passes beyond these into the reflective or ratiocinative. Pure science, then, must be wholly abstract. The forms and substances of Nature with which the scientific student deals, are only the discrete figures of the young mathematician, to be thrown aside with advancing knowledge. Matter is only the staff on which the mind leans, while too feeble to go alone. It is not the finely chiseled statue that renders a man a sculptor; it is the conception which is therein embodied. A day-laborer may have cut the stone, but only the artist could conceive the idea. So in science, we care but little for the particular results at which we arrive, compared with the laws, according to which the results have been attained.

But conceptions cannot be communicated without being rendered objective. The ideal of the artist is locked up in his own mind, until on canvas, in marble, or by means of some other physical symbol, he communicates his high imaginings. Matter, then, according to the present constitution of things is the condition of intellectual communication. Law cannot be studied as abstract law; it can be studied only while acting, and that which exhibits this activity must be matter—something which will always and uniformly obey. There can be no conception of force except as acting, and the sole medium of such activity is matter. Thus again, matter is the condition of all communication from nature to man. Science is thus, in a measure, determined by the conditions of its discovery and communication. But we must distinguish between an invariable condition and that which is thus conditioned. Matter is not science; it is only the condition of its discovery and communication. Air is not hearing; it is the condition of hearing. We do not study matter for the sake of the matter when we study science, but for the sake of the law communicated to us in these changes of matter, and Law is a metaphysical, not a physical idea. Reason, not sense, apprehends it. Law is, so to speak, formulated in the physical, but it is not material. Matter is only the vehicle of science, as language is the vehicle of thought.

It is plain, then, that just as in mathematics we have a division into pure and mixed, according as we deal with matter in the abstract or in the concrete, so we may in any science make a corresponding division, according as we confine our attention to the laws revealed by matter or to the matter revealing the laws: in other words; just as we give attention to the ideas of the message, or to the language in which it is communicated. The language must first be learned, but the words used to communicate the message may be separately understood, and yet the meaning of the message wholly missed. Knowing only the one makes a charlatan; knowing the other makes a savan. The sciences based upon this objective study of Nature are denominated Natural Sciences; and because they lisp the first syllables of Nature's message to man, they should be his primary teachers. It is by their aid that the universal message of God to man must be read. They form, as it were, a public highway leading from Nature to God. But the difficulty is that observing men become so absorbed in admiring some splendid piece of Divine engineering that they stop to gaze and wonder, until losing sight of everything above and beyond, they refuse to advance, fondly imagining that they have reached the end of the journey.

The science based upon this subjective study of Nature is called metaphysics. Logic has been defined as "The Science of Thought;" it should be termed "The Science of Thinking." It is not a dead body which we are studying by dissection, but a living, vital Force, which we study by observing its activities. We find here the same error which we find elsewhere—a stopping with the material symbol, and an ignoring of the intellectual force which clothed itself with the symbol. Astronomy is not the science of circles and spheres, ellipses and ellipsoids, but of the Force whose sensible utterances are given in these curves. We might as well call Painting the science of pictures, or Sculpture the science of statues. So Language, the medium of thought, is only a symbol, less material indeed than pictures and statues, but still physical. What we want in "The Science of Thinking" is not the knowledge of symbols, but the knowledge of that which is symbolized. The chemist does not care for the compounds he finds in his retort; he seeks after the truth which these compounds formulate. Metaphysics and Physics evidently agree in this; that both are seeking to frame an articulate utterance of the Idea given in the diverse manifestations of Force—the Idea which includes all Potencies, the summing up of all phenomena into that final generalization which includes the intellectual as well as the material, until at last we reach the essential unity of all Truth.

Science, then, is classification, or the discovery of the principles of classification, rather than an arbitrary acquaintance with things classified. Every science, however, must have an objective expression—that is, must be formulated. In this, both metaphysical and physical science agree; the only difference in this respect is, that in Physics, Nature gives us in the first place the material interpretation of the idea—that is, the basis of classification—which we have only to translate into idea: while in metaphysics, we first have the idea to which we must furnish the objective utterance. We see here the precise difference between what is called the logical and the natural method—the one being usually called the reverse of the other. The difference is not so much a difference in intellectual procedure as in objective expression. For instance: The botanist has before him the whole range of vegetable forms. He notes resemblances and differences, and groups plants into species and genera, but his work is not ended when these are named and known, and their qualities discovered. He is seeking amidst these multifarious forms for the law of vegetable growth and reproduction. Every organ of the plant is the symbol of an idea, and these ideas form the science of Botany. These Ideas are metaphysical—that is intellectual, and only their sensible manifestation is physical. The symbols of these Ideas, being given in Nature, must be learned from observation before they can be used intelligently, just as words must be learned before one can speak a language. Mastery of the means of expression is as essential to the communication of ideas an is the possession of the ideas themselves. The botanist observes an individual plant, and notes its characteristics. He observes others which possess some of these characteristics whilst others are wanting. He forms a class-type from these agreeing attributes, and gives this new collocation of characteristics a name. Nature never presents this class-type absolutely; it is found nowhere but in intellect. What has the botanist done but to retranslate the communication of Nature into Idea, and then to express this idea by less complicated and less physical symbols? Man's province in this case is simply to interpret the hieroglyphics of Nature into a more readily comprehended language—to express that simply which nature has expressed confusedly. The scientist restricts himself to the interpretation of a single class of symbols, as the Botanist to plants, the Zoologist to animals, but the end sought in each case is the same—that is, to change all these physical utterances of Nature into Idea, and to secure for this Idea a method of expression involving the least possible materiality of symbol—that is, to change individual facts and phenomena into general principles, which, because abstract, are unchangeable. When this has been done, the work of the Naturalist ceases, but the work of Man, the Thinker is not done; it is only just begun. By assuming the ultimate expressions of the various natural sciences as individual and not as typical, we can treat the truths reached by them precisely as the Botanist treated plants, and, rejecting points of of difference, may find in them all some central idea. This is the province of the metaphysician. He seeks the law of Idea, he determines the law of Thinking, just as all other laws are determined, from a study of the symbols formulating its expression in Nature. When this law has been distinctly enunciated, and freed from all intermixture with the contingent, then the work of the metaphysician ceases, the *summum genus* has been reached. The truths communicated in the symbols of Nature, have been correlated and enunciated, and finally translated from the dialect of man the physical into the language of man the intellectual. Physical science determines the separate words of this message of God, the letters of which are scattered throughout Nature. Metaphysics combines these words into propositions which enunciate a distinct truth. There is therefore neither conflict nor variation between the method of Logic and the method of Nature. The movement of both is in the same direction; the only difference is in the point of starting. And another truth no less important, which follows from the foregoing discussion, is that the method of Nature is fundamental to the method of Logic. Physics should precede metaphysics, but not exclude it; both are essential to every true science, and physics, which stops with physics, leads man by dazzling promises into some Utopian desert only to leave him there to die of hunger. And it is no less true that metaphysics, without this basis in experimental science, is illusory and untrustworthy, wherever the original data are necessarily empirical.

Two conditions are thus necessary to all science: a body of knowable truth capable of being systematized; and an intelligence capable of apprehending and systematizing it. One of these conditions is physical and one is metaphysical; and all true science must be the resultant of Law and Idea, the Objective and the Subjective, the twin forces of Nature and Man. If either of these conditions be wanting, there can be no true science, for science can neither be "evolved from the depths of the personal consciousness," nor can the scattered letters of scientific truth, as given in nature, arrange themselves into the words of a significant message. Knowledge must be classified before it is science, and that which classifies can only be intellect—discovering and enunciating this classification according to the laws of mental action. As prominence is given to one or other of these two conditions we have the division into Logical and Natural, but the fundamental principle of classification is the same in both—it being simply the law of intellectual action—just as the law which governs the action of the levers of a loom will determine the pattern of the woven fabric. There can, therefore, be no conflict between the methods of Logic and those of Nature. The determining element in all classification, whether of the phenomena of Mind or of the grosser phenomena of Matter is uniformly and always the same—the law of intellectual action.

Science then resolves itself into a determination of this Law of mental activity, so that in an ultimate analysis, all science is metaphysical, just as all science primarily is physical. Here, as elsewhere, Law can be studied only in its objective manifestations. The Law of Thinking can be educed only from expressed Thought, but the Law is not objective thought, any more than the idea of the sculptor is marble, or the conception of the painter is paint. The simplest expression of thought is not the syllogism but the logical proposition. Now, it is plain that if the proposition is the formulation—the material representative of thought—if we study it as we study other natural symbols, we will find in it the fundamental Law of Thinking, and ultimately the fundamental Law of all Science: just as, if it were possible to reduce all elementary substances to one, the chemist would be able to find in that one a condensed expression of chemical science.

What then is a proposition? Simply stated, it is the assertion of relation between two terms; or more abstractly, it is the reference of an individual to its species—the assertion of a classification. We find here the same duality which we noticed above. If we give prominence to the individual notion, we consider the proposition in extension; if we turn our attention to the specific notion we consider the proposition in intention: in the one case referring to the individuals composing the class, in the other to the attributes composing the class-type. The first corresponds to induction, the second to deduction. When we study individuals we study physics; when we study the attributes composing the class-type, we study metaphysics. The Law of Thinking as educed from a study of the proposition is the law of classification. The proposition, considered affirmatively, asserts explicitly agreement between certain attributes of two terms; that is, it asserts a classification. The aim of science is to reach this proposition, to discover and assert the principle of classification—in other words, to formulate metaphysically what nature has presented physically. We must find, then, the first or fundamental law of thinking in this *integration* or classification. This fundamental law may be subdivided into two species, according to the two terms of the proposition; of which the first may be stated thus: "Every possible object of thought is to a certain extent identical with every other"; and as the proposition implicitly states disagreement, the second may be stated thus: "Every possible object of thought is to a certain extent diverse from every other." The first gives the positive (subjective) condition of the proposition, the second the negative (objective) condition: both together constitute the conditions of thinking. The proposition is thus the assertion of the same in the different. The proposition also asserts, implicitly, the *tertium quid*, or the basis of classification—the class-type, to which both terms are referred—that is, the proposition secondarily asserts an analysis. According to the first condition we have the inductive process; according to the second we have the deductive process. A complete movement of idea from its purely physical symbolization to its metaphysical interpretation, must involve both these processes.

The mind possesses the power of analysis; it can watch its own operations and retrace its steps, until it arrives at the original data of consciousness; but analysis cannot comprise the whole of the logical process. Before there can be analysis there must be something to be analyzed; before steps can be retraced, they must be taken. We must not confound a condition with a Law—the one is a conception antecedent to all action, a genus to which the particular activity may be referred; the other is coincident with action. The one is the medium of the other. We may illustrate this idea by science itself, which is reached only by an analysis of Art. Matter is the condition of the expression of an idea; hence to all but the artist, Art must precede Science, but this cannot be in the case of the artist; in his mind the Idea is first conceived, and there it is given expression in the forms of Art. Here, as uniformly in Nature, the whole absolutely precedes the part—the universal exists before the particular—God before man. Truth absolute thus exists before truth conditioned. Science before Art. Remove conditions and the conditioned becomes the absolute; art and science coincide. But truth which is assumed to be out of all relations, cannot be comprehended by man, and practically is not. Even the universal propositions of deduction express universality under conditions—that is universality of relation; just as infinity in mathematics means that which passes measurement, while in fact between infinity and measurement there is no relation, and the infinite is thus incomprehensible as an object of thought, although by no means unrecognizable as a necessary condition antecedent to all intellectual action. It is of vital importance that we note this distinction, because reasoning, i. e. classification, is possible only so long as we deal with what is admitted to be under relation: if we assume a term to be out of all relation, it ceases to be an object of thought—it can neither be classified nor unclassified; it is beyond reason. Mathematics can proceed with its investigations only so long as it treats all quantities as measurable; it must wholly cease its calculations if an infinite term be introduced. To claim that analysis represents the complete normal action of the intellect in reasoning, is ultimately to claim that the initial point of thinking is the *summum genus* of thought—God. Now God is undoubtedly the initial point of absolute thought, but he is not the beginning of human thought. Intellectually speaking, God is the final generalization; every movement possible to him must be one of analysis—a differentiation of Himself, so to speak, by negatives. Thus the course of absolute Thought, beginning with God, must be first towards a complete differentiation into ultimate individualization; and lastly a complete integration again of individuals into an infinite whole. This dual action completes the circle of intellectual activity. We have dropped attribute after attribute until we have reached the last possible analysis; but we do not stop here, but by the assumption of attributes we again reach the highest possible synthesis. This must be the method of the divine activity, successive differentiation and integration, the closing in of a mighty circle of infinity, embracing all the finite, but never losing the essential characteristic of the infinite.

Now, if this also represent the exact movement of the finite mind in action—that is, in reasoning, man must be God. Man is finite. Even his infinite is only the immeasurable—not that which is without the category of measure. He cannot begin where the Infinite begins, at the highest possible generalization,—but he must begin with the finite. If what we have shown above be true, man must begin with the individual, and the first mental act of the positive character of thinking, is the reference of this individual notion to a class. Now the *class-notion* is the same as the individual notion, less certain attributes as *individual* attributes, but gathered into a larger whole. This process is plainly integration; we are rejecting from the new conception whatever prevents enlarging the class. Each higher generalization involves all the attributes of the lower, not individually, but specifically or generically. In the final generalization, extension and intension coalesce. Just as we reach the individual by differentiating a universal through successive negations, we reach the universal again, by integration, by successively denying the negations through which we just now differentiated. The movement of the finite mind in reasoning is thus from the individual through the universal to the individual again.

Science thus parts into two great branches—one seeking to establish principles by what we have called integration, and the other the elucidation of facts by *a priori* reasoning instead of observation. That is, the aim of true science is to free man from the restrictions of the finite, and to place him in possession of the infinite—the closing in of a lesser circle of infinite truth, yet never losing hold upon the finite. In accordance with this view we see science pursuing its integrations until it has identified as composing an essential unity all the various manifestations of force. This is the finite becoming the infinite, for unity is, in so far, infinity—God is one, a unity, not a unit. But we also see science going beyond this point, and by a new series of differentiations reaching truths new to experience, if indeed not impossible to experience.

Between these two limits all knowledge is forever moving. It can never rest. The tide of thought sweeps onward towards the infinite—God following it to its final absorption into the *I Am*, simple being,—while finite man, because of his finiteness, can only reach those universals which are infinite only to human thought. Like men on a journey we leave the train when we have reached our journey's end, but the train passes on out of sight in the distance, sending back, now and then, tokens of its progress, as it thunders over a bridge, or whistles shrill as it nears some further stopping place, until at last all is still, not because the train has stopped, but because we can follow it no further with our senses. Even after science has reached the utmost limit possible to it, it is not satisfied to rest there, but starts at once upon its return trip, to bring to notice undiscovered facts hidden in these mighty generalizations. Thus the pendulum of intellectual activity unceasingly vibrates between the infinite and the finite, never resting, because Idea and Matter, the force of Man and the force of Nature can never be completely identified.

THE PHILOSOPHY OF EVOLUTION.

The intellectual processes of a rational being must proceed according to some law. They cannot succeed each other at hap-hazard. The notion of rationality is conditioned upon this regular procedure; if this be wanting, the essential character of rational action is wanting. But to say that rational processes are determined by law, and conditioned upon a regular procedure, is simply to assert that the steps in ratiocination are so related to each other that the relation of each to every other may be determined by the application of the law—the difference between any two steps being analogous to the difference between any other two. The astronomer determines the orbit of a planet from three observations, because he thereby determines the law of variation between these points; from which he assumes that this law will be constant, presenting a series of terms each differentiated according the series of differences already determined.

Applying the same principle to mental phenomena, we may determine the law of intellectual action. Thoughts are discriminated by the presence or absence of certain attributes. At one extreme we find the *summum genus*, comprising the fewest possible attributes distinguishing an idea; at the other extreme we find the individual, comprising any number of attributes. Between these two extremes we find a regular series of intermediate terms. The movement of an idea from the general to the individual is like the motion of a planet through one-half of its orbit; while the return movement from the individual to the general, corresponds to the motion of the planet over the remaining half of its orbit. The same law governs both movements and unites the two halves of the orbit into a single whole; and a series of observations taken at equal distances, will, by the uniformity of differences presented, reveal the operation of the same law in this dual manifestation. Upon examining the processes of deduction and induction, we find in each the same series of terms, differing only in the fact that they are in inverse order, and this correspondence reveals the operation of one and the same law. An inductive series is only a deductive series read backward. Any two terms in a series whether inductive or deductive, differ only in the degree of generality, and differ similarly from a third term, so that two being known the third can be therefrom determined. In a deductive series the terms differ by a constant increase in the number of individualizing attributes—a concept being expanded into a deductive series by such regular additions. Having two terms we can proceed to the third—that is, from two propositions expressing this relation, we can proceed to a conclusion. In an inductive series the terms differ by a constant diminution in the number of individualizing attributes—an individual term being expanded into an inductive series, by successively dropping the attributes which compose the individual term, until we reach the required degree of generalization.

Thought must proceed in one of these two directions. The object-matter of thought being composed wholly of attributes can differ only in the presence or absence of certain attributes. A combination, then, of these two movements must complete the intellectual orbit. The direction of the movement of the mind will be determined by the end proposed. When we possess the knowledge of phenomena and wish to discover law—that is, when we seek information—we proceed by induction, from the individual to the general. When possessed of knowledge, we wish to discover its applications, when knowing the law, we wish to determine the phenomena necessarily resulting therefrom, we proceed by deduction—from the general to the individual. Complete knowledge, then, consists in the highest possible generalization, and the expansion of this term into a series, ending only with the last possible individualization. The aim of physical science is to determine that half of the intellectual orbit which lies between the individual and the general—the aim of metaphysical science is to trace the other half which lies between the general and the individual. When we seek to know what is, we proceed by induction—the method of the phenomenal. When, knowing what is, we proceed to determine what hence must be, we proceed by deduction—the method of the Necessary. Thus Science, at first seeking principles, proceeds by induction to establish them; but after these fundamental principles have been established, it proceeds deductively to determine what must result from them, without waiting to discover these truths by observation.

Knowledge is thus complete just in proportion to the extension of its scope through generalization. The higher the generalization, the more inclusive will it be, and the *summum genus*, or the final generalization, will be the highest attainable reach of knowledge. When man can make no further generalization, his knowledge will be, in so far, absolute and complete, and all that remains possible to him will be the practical application of what he already knows. Perfect knowledge is nothing but perfect generalization. The Supreme Intelligence being hypothetically possessed of all knowledge, that is, having discriminated the absolute *summum genus,* can proceed no further in this direction; his intellectual activity must be exerted in a descending series, or from the general towards the individual, and this process must be, as we have seen above, by a determinate series of steps, fixed by the operation of a definite law, which law proceeds by the successive addition of attributes to the general.

Complete knowledge, being complete generalization, the lines of all science will necessarily converge, as they approach this generalization, until all sciences coalesce in one science, and all truth is reduced to a single expression in the utterance of the final conception. In accordance with the laws of thinking, this general term is reached by successive omissions of particularizing attributes, until at last we reach Being—the absolute *summum genus,* wholly free from individual attributes, and thereby embracing everything possible to thought, whether material or immaterial. But this *summum genus* must be predicable of this whole. Matter and mind may thus be reduced to a single category, and the physical and the intellectual finally coalesce in this last generalization. Materialism and idealism thus differ merely in the degree of generalization reached—or rather they both agree in avoiding the final generalization which identifies both matter and mind. Materialism must always deal with the individual, for matter can appear under no other form. Idealism must always rest upon the general, for thought, to be thought, must state a generalization. Each, however, finds its explanation in the other, and both are harmonized by the application of the law of intellectual action above given. Matter and Mind are complementary, not incompatible. They differ with each other, but they agree in being similarly related to a third term. Matter is objective; it is thought taking form, becoming individual, manifesting itself in space. Mind is subjective. The one appeals to the senses; the other is known only to the consciousness.

Science reaches its full development only when it includes both physical and intellectual phenomena within its scope. Every step which it takes carries it further from the purely physical, and brings it nearer the purely intellectual—that is the development of physical science is from the individual towards the general, and it reaches its end, its completion, only when the last distinction, that of subjective and objective, has disappeared in the last possible generalization. When the objective has been identified with the subjective, the distinction between Mind and Matter has been obliterated, and we have reached the Supreme Intelligence—the "I Am" of Scripture—simple Being.

Matter is the formal expression of thought, or the necessary condition of such expression, and in this condition is found the link that connects the subjective and objective manifestations of *being*. Subjectivity is ideality, as objectivity is materiality. The consciousness can take cognizance only of what is within itself, and therefore without every other. Consciousness is therefore wholly personal. To communicate an idea it must be placed within the consciousness of another. To reach this result it must cease to be personal, must pass out of the subjective consciousness into objective form, so as to be placed in the same relation to the speaker and the hearer. Thought, out of the consciousness of the thinker, is objective to him, and to render thought objective is to give it material form. Thought to be communicated, must pass out of the consciousness of the thinker into a material representation. The assumption of material form individualizes the idea. The artist's mind may be filled with splendid conceptions, but no one but he can look within his consciousness and see them. Before others can have any knowledge of his thoughts, he must give them form, or embody them in statues or paintings. The soul of the musician may be thrilled by the harmonies that his imagination creates, but no other soul can join him in this ecstasy until he has given form to his conceptions. So the thinker must embody his thoughts in language before he can communicate them to another. Matter, then, is the vehicle by which thought is communicated, and, so far as we are concerned, the necessary condition of such communication, so that the conception of thought apart from the thinker involves the intervention of material forms, and it is by the interpretation of these symbolical forms that we discover the idea.

Now, let us suppose a Supreme Intelligence. The intellectual processes of such a Being, to be conceived as rational by us, must be identical with ours, or at least analogous to ours. The possession of infinite attributes may in fact free him from the control of any law, but it is impossible for us to conceive an intelligence acting otherwise than in accordance with law. So that if the Supreme Intelligence is to communicate with man, it must be in obedience to the laws which control our mental activities. The Divine thought must, then, like human conceptions, be communicated by means of physical symbols.

The Supreme Intelligence, being the final generalization, must possess all knowledge, and the only intelligent action possible to him from our point of view, is from this absolute generalization towards the concrete and individual. The absolute general is purely subjective, which, to become cognizable, must be rendered objective. This can be secured to us only through the intervention of material forms. From this point of view, matter is only the symbol of thought—thought apart from the thinker. The first result of the divine activity in self-manifestation would be the analysis of *being* into subjective and objective—that is the discrimination of mind and matter, which terms are severally the final generalizations of the two fundamental divisions of science. Matter, then, mere formless, chaotic matter, would be the first result of creative activity. Following the development of this idea in its continually increasing individuality, as new attributes are severally added, matter assumes determinate form and becomes related in systems, as the various so-called elementary substances are discriminated, until finally all truth, capable of being revealed by inorganic matter, is presented to us.

Add the idea of organism and we have the two great divisions of phenomena—material and vital. The higher the generalization, the fewer will be the attributes composing the concept, and thus the simpler will be the form symbolizing its expression. As in the case of matter, the first result of the divine activity was more matter, undiscriminated by any further attribute; so here, we have, as the first organic creation, a concrete expression of the highest possible generalization comprising the fewest possible attributes—that is, forms of life involving the fewest individual characteristics. To matter add the simplest organic attribute—that is, the one lying nearest the genus—and we have mere organized matter, the simple cell, the foundation of all life, no matter how great its future complexity, equally the origin of animal and vegetable growth, which are as yet entirely undiscriminated. This would be the first appearance of life.[1] Differentiating again by the addition of a new attribute, and organic being is subdivided into the two species, vegetable and animal. Beginning with these typical forms, adding single attributes in a continuous series, we at last reach the highest types of animals and plants. Finally, add rationality to the animal, and we reach man, the highest and therefore the most complex type of life, and who, so far as we are concerned, must be the end of creation. We cannot conceive of any higher creation, because we cannot add an attribute to those we already possess, any more than we can conceive of an additional sense by which to cognize such new attribute.

This process has been determined from the very outset by those intellectual laws which we cannot disobey, and which we cannot conceive disobeyed by an intelligent creator. If the law of intellectual action require this process from the simple to the complex, the concrete representation of the steps of this process must indicate the operation of this law, and must also proceed from the simple and rudimentary to the complex and highly developed. An intelligent Creator in revealing his thought must follow the method which our minds must follow in interpreting this revelation. When we know and seek to communicate our knowledge, we proceed from the general to the specific.[2] The Creator assumed to be infinite in knowledge would therefore follow this process instead of the method peculiar to investigation. The law of intellectual action determines this method, and the conditions of intellectual communication determine the representation of this method in the material expression of the ideas communicated. Considering the operation of this law under these conditions, we find that the thought communicating only, as nearly as may be, the generic idea, will be distinguished from it by the addition of but a single attribute as the generic by itself is incapable of being represented in concrete form, the expression of this thought in form will present us matter distinguished from matter in general by but a single attribute. The least possible individualizing attribute added to the highest possible generalization gives us the simplest expression of an idea, and the form or the organism symbolizing this thought will be the simplest form and the simplest organism possible. For instance: in organic life the highest generalization barely individualized will give us the simple cell; and no matter what degree of complexity we subsequently reach by the addition of an almost infinite number of attributes, we nevertheless begin in every case with the same starting point.

Each higher type is reached by adding to a lower. The higher thus embraces all that can be found in the lower, and something besides. This method is invariable, and can never be departed from. The genus must always be predicable of every individual component of every species contained under it. Translating this law into the forms of material expression, and it requires each higher species to physically include all lower species, and to differ from them only by addition. Man, the highest type, must thus include all the attributes of the cell as physically expressed, and without them he would not be man. The differences between no two terms in a series can be total. If the successive steps in a train of thought must be related, so that no two notions will be wholly distinct from each other, these notions will constitute a series, each term of which will, in a measure, determine the next, so soon as the law of the series is discovered; and if this train of thought be objectively presented, it will afford a corresponding series of physical terms, each one of which will in like manner determine the next. But thought is impossible unless by a train of ideas so related. Its physical expression will therefore be equally impossible except by a series of physical terms similarly related, each one of which in some manner determines the next. There must then be a perfect continuity in the line that reaches from the simplest form of matter through all grades of organic life up to man, the highest expression of the divine idea. There can be no break in the chain of thought, because the law of the logical process forbids it: there can be no break in the series of material symbols for the conditions of concrete expression equally forbid it. A symbol is nothing except as it represents that which is to be symbolized. So the symbols form a physical series, because the thoughts symbolized form a logical series.

If the creator has fully revealed his thought, it must be by a series of physical terms arranged in such a manner as to indicate the logical series of ideas symbolized. Every form of matter is a symbol of thought, and challenges interpretation. Every change in form corresponds to an antecedent change in idea, and must be intended to reveal it. As thought, then, begins its evolution with the general and proceeds to the individual by a series of terms each of which is similarly related to both extremes, we must find the material enunciation of this process assuming the form of a series of terms, beginning with mere nebulous matter, grading into organic life, and organic life presenting us with a similar series beginning with the mere cell and ending with man. So rigid and invariable must this serial arrangement be that if a term in either series be wanting, we are authorized to hypothetically interpolate it.

"Nature never makes a leap," says the scientific investigator, as he studies the material symbols of thought. "Thought never makes a leap," says the metaphysician, as he studies the necessary laws of rational action: and both have uttered the same truth. We prove a proposition by determining the steps by which it was educed from a more generic statement. Science must proceed in the same manner, for science only discovers the track of mind—it does not make the track, it only follows it. If then we find the chain of evolution broken at any point, science must either stop there, or assume the wanting term in the series. We have the right to interpolate these missing terms, for we must assume that the thoughts of God communicated to us in material forms constitute a continuous revelation, beginning with Himself, the final generalization, and ending with man the highest individualization. These limits are fixed—the one by the nature of God, and the other by the nature of man. Between these two extremes we must find a series of intermediate terms. Any other conception of their relation than that of a determinate series is impossible and irrational; and a series, so far as it means anything, means evolution of some sort. Finding the relation between these terms—distinguishing the *same* which reproduces itself, and the *different* which introduces a new term—that is, determining the law of apparent evolution—is the problem presented to science.

The astronomer found Bode's law to all appearance violated by the omission of a planet between Mars and Jupiter. He could see no reason for the law, but if the planets had been placed by an intelligent Creator, some order of arrangement must be discoverable according to which their position was determined. The Creator being intelligent, it is impossible to conceive them placed fortuitously. There must then be a link between Mars and Jupiter, because the law once established cannot be broken. The same law may be observed in the arrangement of leaves around the axis of a plant. If intelligence arranged them they must be arranged in some order, for intelligence never performs the least act without a purpose. Each leaf or pair of leaves is not a mere duplication of the previous leaf or pair of leaves. The relation which subsists between any two sets in the series expresses the idea of the Creator, and this must be constant. Completing the series as indicated by different plants, we may assume that if any term is apparently wanting, it is only because it has not been discovered. In neither of these cases would it be asserted that any physical evolution had taken place—the terms form a series of which each term is equally determined by the operation of a fixed law; and yet it is an operation precisely analogous to that which in the case of animals presents every appearance of a real evolution. Take, for instance, a series of animals, presenting at one period of time the simplest and most rudimentary forms, and at another the most complex and highly organized; we cannot do otherwise than conceive these two extremes as related by intermediate terms, through the operation of some law which holds good throughout the series. The relation subsisting between any two, must be the same as that subsisting between any other two similarly situated, or a departure from that relation which is itself governed by a definite law discoverable from a comparison of two sets of terms. The application of this law is so universal and so rigid that we need not hesitate to interpolate a missing term, and confidently assert that it either does exist or has existed. To deny this principle is to deny the necessity of continuity in reasoning. This continuity of thought is represented in matter by the persistence of generic forms under specific differences. But just as the specific is the generic with certain additions, so the individual is this same generic with still further additions; and these additions, whether considered solely in space, as given in the symbols of physical science, or in time as in the conceptions of intellectual science, must be determined by the same unvarying law. The persistence of the same form furnishes us the means of identifying this relation, while the differences reveal to us the successive steps by which the generic was differentiated into the individual.

If the creative thought has been expressed by the forms of matter, the laws of thought must be thus expressed in the relative forms of matter. Anything less than this, while it might interpret isolated ideas, would not communicate the method of the creative process, and science is nothing but the discovery of this method. If the terms of the logical process must be arranged in a series, the physical symbols rendering this logical process cognizable, must be arranged in a similar series, for science becomes impossible when the logical process becomes undiscoverable.

The differences between the terms in this series must be cognizable. Two terms which are indistinguishable are practically identical; and two terms which are not identical vary by a difference which is cognizable by itself apart from either term. The steps in the logical evolution of the final term. *Being* must be separable to be cognizable, and the material forms interpreting these steps to the senses must also be distinguishable. A species differs from the genus by the addition of at least one attribute. Now, if the species is distinguishable from the genus, the attribute which differentiates it, must be separately cognizable—so also the individual differs from the species by the addition of attributes, which must in like manner be separately cognizable, or the species will never be conceived independently of the individuals. A thought cannot proceed by insensible steps, nor can its material expression vary otherwise than by determinate and distinguishable differences. The distinction of species is thus a logical necessity. The addition of distinct attributes to the genus gives origin to distinct species; variation in attributes not affecting their substantial identity gives rise to varieties. One species, then, cannot become another, except by the assumption of a new specific attribute, so that one species passes into another precisely as the genus passes into the species, and that is just as, and not otherwise, than one thought passes into another.

The fundamental law of the logical process is that we pass from the generic towards the individual; from the simple to the complex. Induction can proceed only by assuming a genus at the outset—that is, by assuming certain attributes in the individual to be generic. Translate this law into material forms, and we have each higher—that is more complex—species evolved from the lower by the addition of some new characteristic. This new attribute cannot be added by the functional activity of the lower organism; that can only reproduce itself. A thought does not change merely through repeated expression. We pass to the conclusion of a syllogism, not from each term, but from a comparison of the premises—and this requires an intellectual operation entirely distinct from a mere apprehension of the terms. It is one thing to comprehend the premises; it is quite another to deduce a conclusion from them. It may necessarily follow, but it requires a separate act of the mind to reach it. Premises will not of themselves reach a conclusion.

Reading this same truth in the forms of matter, we may say that species will not pass into higher species without the intervention of a force distinct from either. The impulse which adds a new attribute must be intellectually separable from all those pre-existing, and its material representation must be physically distinct from pre-existing forms. This complete separability precludes the possibility of mere physical genesis. The added attribute is presented by a new form of matter, revealing the presence of a new thought—a new effect, requiring the agency of a new cause. In accordance with the usual economy of nature, who never duplicates her forces, change will be made only so far as may be necessary to communicate the additional idea. Organisms representing previous thoughts will be added to, in order to express the expansion of the thought, instead of a creation *de novo* in each instance. Thus an identical cellular structure will be found in all organic beings, from the lowest to the highest, each higher type carrying forward the idea and its physical expression found in the lower. The differences between no two terms in the series can be total, nor can any two terms be identical, as each higher species will embrace all the attributes of the lower, differing only by the addition of others. This is simply the physical expression of the logical truth that whatever can be predicated of the genus can be predicated of every individual contained under it. As the individual is only the expansion of the genus, so higher physical types must also be similar expansions of lower.

Here, then, is evolution, or development: primarily an evolution of the generic into the individual, the continued differentiation of a generic idea through successive individualizations, each adding to the previous group of attributes, thus rendering the idea increasingly complex; and, secondly, an apparent physical evolution or development, interpreting this logical process by a series of physical forms so related as to reveal the relation existing between the thoughts thus interpreted. In the physical representation of the ideas so related, there must be an apparent physical evolution—that is, the process of evolution logically must, like the ideas thus evolved, have a physical expression, and the successive steps in this logical evolution must be revealed by material forms bearing an analogous relation, and thereby expressing the logical process. Matter is nothing, so far as we are now concerned, but the condition necessary to the objective expression of thought. Every phase of matter is simply an objective formulation of a corresponding phase of thought. Every addition to form implies an antecedent increase of thought, as there can be no formal expression until there is something to be expressed. There can, then, be no such thing as mere material evolution, for whatever is material is only symbolical.

Matter being thus wholly inert, the origin of the impulse towards greater complexity must be sought for outside of that which undergoes the change. The movement by which one species becomes a higher is not an elaboration, an extension or a differentiation of existing attributes, but involves the positive addition of a new attribute, different and distinct from any or all previously existing. One species cannot pass into another by an innate impulse, for a species is an entity composed of a determinate number of attributes, and all attributes potentially present must be considered as actually present. We cannot say that the child is a different species from the man, and that one passes into the other by a process of evolution, because all the essential attributes of the man are potentially present in the child. If the polyp, by the action of innate forces, operating through a series of ages, however extended, can, without any impulse from without, develop itself into a man, then the polyp is as much a man as a boy is, differing only in the time required for development: and the data for the final deduction of the highest types of creation must be furnished in the most elementary forms of life.

The force manifesting itself in organic life is readily distinguishable from the organism by which it is manifested. Life and organization are not synonyms; one is the condition of the other, but a condition is not a cause. We can consider force apart from organism, and this possible separation in thought proves that the same form may not represent both, but that life can absolutely exist apart from organs which serve to give it a physical manifestation.[3] Physical life being conditioned upon organization, whenever the organism varies, the vital force thus manifested must also vary, such variation being necessarily antecedent to its manifestation. The organism varies, because it must, in order to express the added thought. Change in organism, therefore, is not induced by simple organic action, because the organs and the force acting through them can be distinguished. Assuming that matter is the objective or formal representation of thought, there can be no change in the material expression without a corresponding change in the antecedent conception. There can, then, be physical evolution, only as there is antecedent logical evolution, and then only because of this logical evolution and not because of the operation of an innate organic force. Force, whatever may be its genesis, is only the exertion of power, not the increase of it. Exertion limits the view to the force immediately in operation. We may replace one manifestation by another, but the quantity is neither increased nor diminished by this change. Change in form implies the operation of force: and apart from such manifestation in matter, it escapes the tests of science, and passes into the purely metaphysical notion of cause. And unless the operation of force be constant, or, if different forces are blended, variable according to some determinate law, the action of which is constant and discoverable, so that the different units of force are separately measurable, the force thus irregular in its action can never be placed in any scientific category. Evolution, then, cannot proceed from any innate organic impulse, unless the force that tends to exact reproduction, and the force that induces a change be equally and separately cognizable. Change must proceed according to some law which accounts for the change, and distinguishes between the normal exertion of power and that exertion which causes a deviation. Science, to be science, must explain apparent exceptions as fully as the regular operation of forces, and that which causes the irregularity must be as distinctly cognizably by itself as the force which acts regularly. Anything less than this is not science. The discovery of Neptune was the result of the application of this principle; it was a successful attempt to discriminate the force which caused variation from the force which operated regularly.

Each species represents the operation of certain vital forces, and one cannot physically pass into another except by the increase of this force, or at least by a change in the manner of its manifestation; and this increase in amount or this change in direction must separately be accounted for. Nor does it matter, for the purposes of this discussion, as to the genesis of this added increment, further than to show that its origin must be exterior to the organism by which its presence is manifested; for vital energy acting through an organism is a unit, and cannot, even in thought, be separated into distinguishable portions. Change in the direction of vital energy indicates that the original impulse has been modified in its action by encountering another force, for nothing but force can change the direction of force. It does not fall within the range of this paper to determine the nature of this exterior force which is thus distinguishable from that acting through the vital organization, and therefore capable of separate objective representation. Metaphysically we may say that force is resolvable into will, but will being purely personal is incapable of material representation, and thus cannot enter into the determinations of physical science, which does not seek to discover the origin of force, but deals solely with its presence.

As the logician must assume his premises, and, as a logician, cannot question their truth, so the physicist must assume a force in operation, and, as a physicist, cannot examine its genesis. The physical or the metaphysical method of inquiry is valid only so long as restricted to physical or metaphysical processes: a mixture of the two methods will give results satisfactory neither to science nor to philosophy. As logic furnishes no criterion by which to test the absolute truth of propositions, but deals wholly with conclusions drawn from given premises, so science furnishes no data by which to determine the absolute genesis of force, but restricts its enquiries to the phenomena resulting from a force given. For the student of physical science cause and effect is only the transference of a given and determined force from one material form to another. If this idea is to be traced further, it must be studied outside the limits of physics. This study belongs to metaphysics.

Now, if physical science does not deal with the origin of the initial force, but assumes at the outset its presence, no more does it fall within its province to examine into the origin of the increments which give to physical forms that variety which renders science possible. Science deals with results, not antecedents; and after having determined results, it is not authorized to affirm that one species has produced another by evolution, or has produced it at all. If there are agreements between different organisms by which they are brought into relation, there are also differences by which they are discriminated, and these differences imply increments of force; and to assert that one organism has evolved another is to determine not merely the presence of this new increment, but also to determine its origin. Scientific investigation deals with phenomena which give evidence to the senses of a *transference* of force from one form or from one manifestation to another. Transference is not increase—an effect can be no more than the evolution of what was potentially present in the cause; it cannot add to it. The origin of the force must be investigated according to intellectual laws.

It has been argued that a Supreme Intelligence in manifesting his thought will, according to the necessary laws of rational activity, pass from the universal and general to the particular and individual, or from concepts involving few attributes to those involving these and others; and that these steps in the rational process must be represented in a corresponding physical series; and that the communication of thought is conditioned upon this physical representation. If the logical series comprises one thousand terms, each related to the preceding according to logical law the physical series must comprise one thousand terms, each physically related in such a manner as to reveal this law. As the highest generalization comprises the fewest attributes, the concrete expression of this idea will present the simplest possible physical form and the least complexity of organization, and thus will present the lowest types of life; and as the individual comprises the greatest number of attributes, its concrete expression will present the greatest complexity, and consequently the highest type of life. We have seen that the logical process begins with the general and ends with the individual; its material expression must therefore begin with the lowest orders and end with the highest. But the individual cannot be immediately derived from the general without the intervention of intermediate generalizations. No more in the concrete expression of this deduction can we pass from the lowest types to the highest without the intervention of an intermediate series. These intermediate terms are not capable of independent interpretation; they find their full explanation only in the extremes of the series—God and Man.

If, then, in the intellectual process from the abstract and universal towards the concrete and individual, we find a constant evolution of idea, each advance being an addition to the previous conception, each new term in the series embracing all the attributes of the preceding, and differing only by addition; and if thought is possible only on this condition; it necessarily follows that the material representation of this thought must present physical forms similarly related, so that, leaving out of view the intellectual genesis of this relation, the observer might conclude that these forms compose a series evolved from a primordial cell in accordance with an organic law. But such we find to be the universal law of intellectual procedure: this apparent development or evolution must, therefore, be the condition of the communication of such intellectual process, and the physical terms are brought into this relation by the fact that they symbolize the logical process. If the material symbols of thought were unrelated physically, the thoughts thus expressed would also be unrelated and independent. But such a supposition readers Science impossible, for its one aim is to find the *same* in the *different*. If there be no *same*, there can be no science: if there be no *different*, there can be no science. Thought proceeds by adding the *different* to the *same* in an endless series, and this addition of the *different* to the *same* expressed in concrete forms is what is called evolution. If no evolution were apparent in Nature, there could be no Science; for those steps which to the naturalist indicate evolution, being only the physical expression—the formulation—of the logical process, afford the means by which the student reaches the highest generalization. If these steps be wanting, he cannot proceed.

Admitting then to its fullest extent the fact that, judged from a purely physical point of view, all organic forms seem to have been derived each from its immediate predecessor, by a mere functional impulse; and admitting that science is possible upon no other condition; we claim that these material forms are brought into such relation by intellectual evolution, and not by physical genesis; they represent an evolution of Thought and not an evolution of Matter. We know from consciousness that this process of evolution is the method of our thinking. We know also that the divine thought can be rendered intelligible to us upon no other hypothesis than that which supposes it to be governed by the laws which control human thought. Translating the physical symbols which we see about us, and which present this appearance of evolution, we infer that this is the method according to which the divine mind proceeded. Science will not materially err in its physical results, if it adopt the hypothesis of physical evolution, but it must confine its attention to physics; it is only as we attempt higher generalizations that the insufficiency of the hypothesis becomes manifest in its failure to satisfy the conditions of the problem as presented to philosophy.

FOOTNOTES:

[1] This, of course, does not absolutely determine the order of organic creation; as in the case of the syllogism the conclusion or either premise may be the proposition first enunciated, the order of expression being determined by circumstances.

[2] Compare the demonstrations of Geometry.

[3] As in the case of man after the death of the body.

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